

A PRECISION THRESHOLD FOR INITIAL-SET DEPENDENCE IN CONTRACTING CONTROL SYSTEMS

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ABSTRACT. We study the smallest control catalogue that keeps each actual initial state within a prescribed Euclidean neighbourhood of a constraint set throughout a finite horizon. For a class of smooth, globally invertible planar systems, both the feasible angular branches and the radial contraction depend on the state. We prove a sharp threshold $\gamma_* = \beta[-a_0 \log a_0 - a_1 \log a_1]$ for precision sequences $\delta_n = \exp(-\gamma n + o(n))$. Below and at this threshold, every fixed positive-area compact initial set has cost rate γ/β . Above it, one compact set of arbitrarily nearly full area has rate $-a_0 \log a_0 - a_1 \log a_1$ simultaneously for every finite precision exponent above the threshold, whereas the full constraint set has a strictly larger lower rate. The comparison with actual approximate controls follows from fibrewise first-deviation estimates and a stopping-witness argument with only polynomial catalogue loss. The nonlinear systems admit no simultaneous conjugacy to the matched linear model that preserves the constraint set. A smooth noninjective example, identical to that linear model near the common attracting origin, shows why local matching and uniform convergence alone do not imply the threshold theorem.

1. INTRODUCTION

Invariance entropy measures the exponential growth of the number of controls needed to keep an initial set inside a controlled invariant set; see Kawan [7]. At a fixed positive tolerance, a contracting system can have bounded outer invariance complexity. This does not determine the cost when the tolerance decreases with the horizon. The latter problem compares two scales: the information needed to choose feasible branches and the contraction that eventually makes further precision constraints inexpensive.

We consider a catalogue of open-loop inputs from which a control is assigned to each actual initial state. Every time from 0 through n must satisfy the same tolerance δ_n ; approximate trajectories may leave and reenter the constraint set. Our question is whether the exponential catalogue cost is independent of the initial compact set as long as that set has positive area. We identify the exact range of precision exponents for which this independence holds in a class with matched angular geometry and radial contraction. Beyond the threshold, removing a set of arbitrarily small area changes the cost strictly.

The main result includes an explicit state-dependent, globally invertible smooth family. Its feasible angular intervals vary with radius, and its assigned digits are not assumed to be independent under area measure. We derive uniform cylinder comparisons from the maps, then establish estimates for *arbitrary* approximate inputs. These steps connect classical word counts to the optimal catalogue of actual controls. They also show that the same nearly full-area compact set works for all higher precision exponents and all precision sequences with those exponents. A boundary-based obstruction rules out a simultaneous change of coordinates preserving the constraint set. A complementary example has the same local dynamics but a global angular collapse, and loses initial-set independence at every positive precision exponent.

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1.1. Relation to previous work. Our finite-horizon counting function is the outer invariance complexity used by Chen and Zhong [1, Section 2.2], with a one-step change of time convention. Their bounded-complexity and equi-invariability results concern each fixed tolerance. Entropies for exponential and practical stabilization [3, 4] also relate control information to accuracy, but impose stabilization tasks different from the all-times Euclidean neighbourhood constraint used here. These frameworks provide the context for the problem, rather than the catalogue comparison proved below.

Measure-theoretic invariance entropies and variational principles have been developed in [9, 8], and invariance entropy dimensions for zero-entropy systems are studied in [2]. We evaluate outer complexity along a specified exponential Euclidean precision scale and compare actual positive-area compact initial sets. The measure in our proof is ordinary planar area, used to construct typical subsets; it is not asserted to be an invariant measure of a single controlled map.

Stopping at a product threshold, type counting, and typical-set arguments are classical tools; see [6, 5]. Yang, Chen, Yang and Zhou [10] establish induced-pressure and Bowen-equation results on a given invariant partition. In particular, their spanning/separated comparison is made for the addresses of that partition. Such a comparison does not itself identify a prescribed exact code with an optimal catalogue allowing approximate controls. Our proof supplies the required controlled geometric comparison: a wrong first branch confines the covered initial states to a thin, radius-dependent band, and one arbitrary input covers only $O(n)$ stopped midpoint witnesses. The resulting threshold is specific to the matching structure stated below. We do not claim a classification of general contracting control systems.

2. THE COUNTING PROBLEM AND THE THRESHOLD THEOREM

Write

$$Q = \{(s, z) \in \mathbb{R}^2 : 0 \leq s \leq 1, |z| \leq s\}, \quad \text{area}(Q) = 1.$$

For two maps $G_0, G_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, an input is $u = (u_1, u_2, \dots) \in \{0, 1\}^{\mathbb{N}}$. Set $G_u^0 x = x$ and $G_u^k x = G_{u_k} \circ \dots \circ G_{u_1} x$. For $\delta > 0$, let $Q^\delta = \{x : \text{dist}(x, Q) < \delta\}$, with Euclidean distance.

Definition 2.1. For a nonempty compact $K \subset Q$, let $r_G(n, \delta; K)$ be the least cardinality of a finite input catalogue $\mathcal{U}$ such that

$$(2.1) \quad \forall x \in K \ \exists u \in \mathcal{U} \ \forall k \in \{0, \dots, n\} : \quad G_u^k x \in Q^\delta.$$

The assignment may depend on the actual state x . There is no restriction on a trajectory between distinct integer times, and leaving or reentering Q at those times is allowed as long as (2.1) holds. Only the first n input symbols matter.

All logarithms are natural. Fix parameters

$$(2.2) \quad \begin{aligned} a_0, a_1 &\in (0, 1), \quad a_0 + a_1 = 1, \quad a_0 \neq a_1, \quad \beta > 1, \\ 0 \leq \varepsilon &< \frac{a_{\min}}{4(\beta + 1)}, \quad a_{\min} = \min(a_0, a_1). \end{aligned}$$

On the entire real line define

$$(2.3) \quad p_0(s) = a_0 + \varepsilon \frac{s}{1 + s^2}, \quad p_1(s) = a_1 - \varepsilon \frac{s}{1 + s^2}.$$

Our maps are

$$(2.4) \quad G_0(s, z) = (p_0(s)^\beta s, p_0(s)^{\beta-1}[z + p_1(s)s]),$$

$$(2.5) \quad G_1(s, z) = (p_1(s)^\beta s, p_1(s)^{\beta-1}[z - p_0(s)s]).$$

The value $\varepsilon = 0$ gives the matched linear maps A_0, A_1 . For $\varepsilon > 0$ the feasible branches and the radial factors depend on s , while $DG_i(0) = A_i$ as matrices.

Use the functions

$$(2.6) \quad H(t) = -t \log t - (1-t) \log(1-t), \quad \chi(t) = -t \log a_0 - (1-t) \log a_1, \quad h = H(a_0),$$

with $0 \log 0 = 0$.

Theorem 2.2 (Common-cost threshold). *For every system (2.2)–(2.5), both maps are C^∞ diffeomorphisms of $\mathbb{R}^2$, Q is controlled invariant, and every input drives bounded sets uniformly and exponentially to the origin.*

For every fixed compact $K \subset Q$ with $\text{area}(K) > 0$, every finite $\gamma \geq 0$, and every positive sequence $\delta_n \rightarrow 0$ with $-\log \delta_n/n \rightarrow \gamma$,

$$(2.7) \quad \min\{h, \gamma/\beta\} \leq \liminf_{n \rightarrow \infty} \frac{\log r_G(n, \delta_n; K)}{n} \leq \limsup_{n \rightarrow \infty} \frac{\log r_G(n, \delta_n; K)}{n} \leq \min\{\log 2, \gamma/\beta\}.$$

Consequently, for $0 \leq \gamma \leq \beta h$ the limit exists and equals γ/β for every such K .

For each $\zeta \in (0, 1)$ there is one compact set $K_\zeta \subset Q$ with $\text{area}(K_\zeta) > 1 - \zeta$, chosen independently of γ and of the precision sequence, such that for every finite $\gamma > \beta h$ and every sequence as above,

$$(2.8) \quad \lim_{n \rightarrow \infty} \frac{\log r_G(n, \delta_n; K_\zeta)}{n} = h < \liminf_{n \rightarrow \infty} \frac{\log r_G(n, \delta_n; Q)}{n}.$$

Thus βh is the exact boundary for the property that *all* positive-area compact initial sets share a cost. The assertion above the boundary is a strict separation, not a two-value classification of initial sets. The theorem does not give the general high-precision exact rate or limit for Q .

The matching relation is instantaneous: on a fibre with radius s , the inverse angular branches have lengths $p_i(s)$ and the radial factor is $p_i(s)^\beta$. It is stronger than a first-order condition at the attracting origin. Sections 6 and 7 respectively distinguish the state-dependent theorem from a constraint-preserving coordinate reduction and from a statement based solely on local matching.

3. SMOOTH DYNAMICS AND COMPLETE FEASIBLE CYLINDERS

Set $a_{\max} = \max(a_0, a_1)$ and

$$\underline{p} = a_{\min} - \varepsilon/2 > 0, \quad \bar{p} = a_{\max} + \varepsilon/2 < 1, \quad r = \bar{p}^\beta < 1.$$

Lemma 3.1. *The maps in (2.4)–(2.5) are global smooth diffeomorphisms. There are $M \geq 1$ and $L < 1$ such that the norm $\|(s, z)\|_* = \max\{M|s|, |z|\}$ satisfies*

$$(3.1) \quad \|G_i(s, z)\|_* \leq L\|(s, z)\|_* \quad (i = 0, 1, (s, z) \in \mathbb{R}^2).$$

For $s > 0$, writing $y = z/s$ and $\xi(s) = p_0(s) - p_1(s)$, the complete one-step feasible intervals are

$$(3.2) \quad I_0(s) = [-1, \xi(s)], \quad I_1(s) = [\xi(s), 1].$$

Proof. For $f(s) = s/(1 + s^2)$ one has $|f| \leq 1/2$, $|f'| \leq 1$, and $|sf'| \leq 1/2$. The radial maps $g_i(s) = sp_i(s)^\beta$ therefore satisfy

$$g'_i(s) = p_i(s)^{\beta-1} [p_i(s) + \beta s p'_i(s)] > 0,$$

because the bracket is at least $a_{\min} - (\beta + 1)\varepsilon/2 > 0$. Moreover it is bounded above by $a_{\max} + (\beta + 1)\varepsilon/2 < 1$. Since $p_i(s) \rightarrow a_i$ as $|s| \rightarrow \infty$, g_i maps $\mathbb{R}$ onto $\mathbb{R}$. Recovering s from the first coordinate and then z from its affine second coordinate gives a smooth global inverse. The Jacobian is

$$\det DG_i(s, z) = p_i(s)^{2\beta-2} [p_i(s) + \beta s p'_i(s)] > 0.$$

Choose $M \geq 1$ so large that $\bar{p}^{\beta-1} + \bar{p}^\beta/M < 1$. The first coordinate in the adapted norm has factor at most $\bar{p}^\beta$, and the second satisfies

$$|z'| \leq \bar{p}^{\beta-1}|z| + \bar{p}^\beta|s|.$$

This proves (3.1), uniformly on the whole plane.

For positive s , the angular maps and their inverses are

$$(3.3) \quad \begin{aligned} T_{0,s}(y) &= (y + p_1(s))/p_0(s), & B_{0,s}(y) &= p_0(s)y - p_1(s), \\ T_{1,s}(y) &= (y - p_0(s))/p_1(s), & B_{1,s}(y) &= p_1(s)y + p_0(s). \end{aligned}$$

Their inverse images of $[-1, 1]$ are precisely (3.2), and their union is $[-1, 1]$. The radial coordinate remains in $[0, 1]$ from Q ; the origin is fixed. Thus every state has an exact feasible continuation in Q . $\square$

For a word $w = w_1 \cdots w_l$ and initial radius $s \in [0, 1]$, write $s_0 = s$, $s_j = g_{w_j}(s_{j-1})$, and set

$$(3.4) \quad P_w(s) = \prod_{j=1}^l p_{w_j}(s_{j-1}), \quad \lambda_w = \prod_{j=1}^l a_{w_j}.$$

Empty products are one. The radius depends only on s and the word, not on the initial angle.

Lemma 3.2 (Uniform product comparison). *There is a constant $D \geq 1$, depending only on the system, such that for all words and all $s \in [0, 1]$,*

$$(3.5) \quad D^{-1}\lambda_w \leq P_w(s) \leq D\lambda_w, \quad s_l = sP_w(s)^\beta \leq r^l s.$$

Proof. The radial identity and bound follow directly from the definitions. For $s_j \in [0, 1]$,

$$|\log p_i(s_j) - \log a_i| \leq \frac{\varepsilon s_j}{p} \leq \frac{\varepsilon r^j}{p}.$$

Summation gives $|\log P_w(s) - \log \lambda_w| \leq \varepsilon/[p(1-r)]$. Take D to be the exponential of this upper bound. Thus bounded cumulative distortion is derived from the maps, rather than imposed as an assumption. $\square$

For each word let $I_w(s)$ be the closed interval of initial angles for which all its intermediate states belong to Q . Set $I_\emptyset(s) = [-1, 1]$. Directly from (3.3),

$$(3.6) \quad I_{iw}(s) = B_{i,s}(I_w(g_i(s))), \quad |I_w(s)| = 2P_w(s).$$

The first formula includes every intermediate constraint, since $B_{i,s}([-1, 1])$ is the full feasible first-branch interval. At each depth the intervals cover $[-1, 1]$ with disjoint interiors. The same holds for any finite complete prefix-free tree of words. Shared endpoints are retained in both closed intervals.

The actual feasible region of w is the curved strip

$$Q_w = \{(s, sy) : 0 < s \leq 1, y \in I_w(s)\} \cup \{(0, 0)\}.$$

The change $(s, y) \mapsto (s, sy)$ has Jacobian s , so

$$(3.7) \quad \text{area}(Q_w) = \int_0^1 2sP_w(s) ds, \quad D^{-1}\lambda_w \leq \text{area}(Q_w) \leq D\lambda_w.$$

For each finite word its endpoint curves are continuous, and have area zero by fibrewise integration. Their countable union and the vertex are a null set, but remain part of every catalogue used to cover Q .

4. ARBITRARY APPROXIMATE CONTROLS AND STOPPING WITNESSES

Assign a natural exact code by taking branch 0 at $y \leq \xi(s)$ and branch 1 at $y > \xi(s)$, and iterating this rule. Denote its length- k prefix at x by $c_k(x)$. This convention fixes boundary itineraries. It is a tool for comparison, not a restriction on the catalogue in Definition 2.1.

For an arbitrary word u of length n , let

$$V(n, \delta, u) = \{x \in Q : G_u^k x \in Q^\delta \text{ for every } 0 \leq k \leq n\}.$$

If a state $x = (s, sy)$ first disagrees with u at time j , let $a = u_1 \cdots u_{j-1}$ and $i = u_j$. The common angular composition $T_{a,s}$ is increasing and affine, with slope $P_a(s)^{-1}$. Put

$$b_a(s) = T_{a,s}^{-1}(\xi(s_{j-1})).$$

At the wrong branch one has the exact identities

$$(4.1) \quad |y_j| - 1 = \frac{|y - b_a(s)|}{P_{ai}(s)}, \quad s_j = sP_{ai}(s)^\beta.$$

At a branch tie the excess is zero; the formula still applies.

Lemma 4.1 (First-deviation band). *The states of $V(n, \delta, u)$ with first disagreement at j lie in a one-sided fibrewise band*

$$(4.2) \quad |y - b_a(s)| < \frac{2\delta}{sP_{ai}(s)^{\beta-1}} \leq \frac{C_b \delta \lambda_a^{1-\beta}}{s}, \quad C_b = 2D^{\beta-1} a_{\min}^{1-\beta}.$$

Its area is at most $C_b \delta \lambda_a^{1-\beta}$.

Proof. If $(s_j, z_j) \in Q^\delta$, the Lipschitz function $(s, z) \mapsto |z| - s$ gives $|z_j| - s_j < \sqrt{2}\delta < 2\delta$. Combine this necessary condition with (4.1) and (3.5). The side of the band is fixed by the chosen wrong branch. Multiplying its angle width by the physical Jacobian s and integrating over $0 < s \leq 1$ gives the claimed area bound. The centre may depend on s ; the fibrewise estimate remains valid. No lower bound on the initial radius is needed. Any later reentry into Q still has to satisfy this first deviation constraint. $\square$

For $0 < t < 1$, let $\mathcal{L}(n, t)$ be the finite complete prefix-free family obtained by stopping at the first word with $\lambda_w \leq t$, or at depth n if no earlier crossing has occurred. Put $N(n, t) = \#\mathcal{L}(n, t)$. Every leaf has

$$(4.3) \quad \lambda_{w^-} > t, \quad \lambda_w > a_{\min} t, \quad \sum_{w \in \mathcal{L}(n, t)} \lambda_w = 1.$$

Here w^- is its parent; the parent of a one-letter word is the empty word. The last identity follows from $a_0 + a_1 = 1$.

Lemma 4.2 (Comparison with the actual catalogue). *With*

$$C_0 = 2\sqrt{2}MD^\beta, \quad B_0 = \left\lceil 1 + D^\beta a_{\min}^{-\beta} \right\rceil,$$

for every $n \geq 1$ and $0 < \delta < 1$,

$$(4.4) \quad \frac{N(n, \delta^{1/\beta})}{1 + B_0 n} \leq r_G(n, \delta; Q) \leq N(n, (\delta/C_0)^{1/\beta}) \leq C_U \delta^{-1/\beta}, \quad C_U = a_{\min}^{-1} C_0^{1/\beta}.$$

Also $r_G(n, \delta; Q) \leq 2^n$.

Proof. For the upper bound, use one input with prefix w for each leaf at $t = (\delta/C_0)^{1/\beta}$, followed by a fixed infinite tail. At every initial radius, the intervals $I_w(s)$ form a complete cover of all angles, including their endpoints. A depth- n leaf is exactly feasible throughout the required horizon. At an earlier leaf, the endpoint obeys $s_l \leq D^\beta t^\beta$ and $|z_l| \leq s_l$, so

$$\|(s_l, z_l)\|_* \leq MD^\beta t^\beta = \delta/(2\sqrt{2}).$$

By (3.1), all later states under the fixed tail have Euclidean norm at most $\delta/2$. They therefore lie in Q^δ at every remaining time. The vertex is fixed. Equation (4.3) gives $N(n, t) \leq (a_{\min} t)^{-1}$. All length- n words give the additional 2^n upper bound.

For the lower bound set $t = \delta^{1/\beta}$. For each leaf choose the actual state $(1, y_w)$ with y_w the midpoint of $I_w(1)$. By (3.6), (3.5) and (4.3), every such interval has length greater than $2D^{-1}a_{\min}t$. Consecutive midpoints have at least this spacing because the intervals have disjoint interiors.

Fix any arbitrary length- n input u . At most one leaf is a prefix of u . Every other covered midpoint witness first disagrees at a time $j \leq |w|$, with common prefix $a = u|_{j-1}$. Since this occurs before stopping, $\lambda_a > t$. At radius one, Lemma 4.1 restricts its initial angle to one band of length at most $C_b t$. Its centre and side depend only on u and j . The midpoint spacing bounds the number of witnesses in that band by $1 + D^\beta a_{\min}^{-\beta} \leq B_0$. Summing over $j \leq n$, one arbitrary input covers at most $1 + B_0 n$ witnesses. A catalogue spanning all of Q must cover all these actual states, proving the lower bound. $\square$

The lower estimate allows every approximate input and every later reentry. It does not assume that the natural stopping tree is an optimal catalogue. The loss is polynomial in the horizon and hence does not affect its exponential rate.

5. AREA-TYPICAL INITIAL STATES AND PROOF OF THE THRESHOLD

Lemma 5.1 (Physical typical sets). *For area-almost every $x \in Q$, the frequency of zero symbols in $c_k(x)$ tends to a_0 . Every compact $K \subset Q$ of positive area contains a compact positive-area set T on which this convergence is uniform. For every $0 < \eta < h$, there is $D_\eta \geq 1$ such that*

$$(5.1) \quad D_\eta^{-1} e^{-(h+\eta)k} \leq \lambda_{c_k(x)} \leq D_\eta e^{-(h-\eta)k} \quad (x \in T, k \geq 0).$$

For each $\zeta \in (0, 1)$ such a compact T can be chosen with $\text{area}(T) > 1 - \zeta$ inside Q .

Proof. Normalize area on Q , whose area is one. For every assigned word w , (3.7) bounds its probability by $D\lambda_w$. Thus the area of states whose length- k prefix has zero frequency at distance at least η from a_0 is at most

$$(5.2) \quad D(k+1)e^{-2\eta^2 k}.$$

Indeed, for $t = j/k$, put $D_{\text{KL}}(t||a_0) = t \log(t/a_0) + (1-t) \log((1-t)/a_1)$. Then

$$\binom{k}{j} a_0^j a_1^{k-j} \leq e^{-k D_{\text{KL}}(t||a_0)}, \quad D_{\text{KL}}(t||a_0) \geq 2(t - a_0)^2.$$

The second inequality follows from $D'_{\text{KL}}(t) = 1/[t(1-t)] \geq 4$, with its value and first derivative zero at a_0 . Summing over types gives (5.2). Borel–Cantelli, followed by a countable sequence of positive η tending to zero, proves almost everywhere convergence. No independence of the state-dependent digits has been used.

Remove the null set of finite-word boundary curves and the vertex. Egorov’s theorem and inner regularity of area give a compact subset of any positive-area K with positive area and uniform frequency convergence. On Q they give a compact set of area greater than $1 - \zeta$. Since $-\log \lambda_{c_k(x)}/k$ is χ of the zero frequency, it converges uniformly to $\chi(a_0) = h$ on each chosen set. Absorb the finitely many early times into D_η to obtain (5.1). All sets are chosen before any precision exponent or sequence. $\square$

Lemma 5.2 (Area covered by one arbitrary input). *If T satisfies (5.1), then for each $0 < \eta < h$ there is a constant C_η such that, uniformly in u, n, δ and $0 \leq m \leq n$,*

$$(5.3) \quad \text{area}(T \cap V(n, \delta, u)) \leq C_\eta \left[e^{-(h-\eta)m} + \delta e^{(\beta-1)(h+\eta)m} \right].$$

Proof. Agreement through m confines the state to $Q_{u|m}$. If this agreement class meets T , its reference product is at most $D_\eta e^{-(h-\eta)m}$, and (3.7) gives the first term.

For a first disagreement at $j \leq m$, a nonempty class in T has common prefix $a = u|_{j-1} = c_{j-1}(x)$ for some $x \in T$. Equation (5.1) bounds its product from below by $D_\eta^{-1} e^{-(h+\eta)(j-1)}$. Lemma 4.1 bounds its area by $C_b D_\eta^{\beta-1} \delta e^{(\beta-1)(h+\eta)(j-1)}$. The sum over $j \leq m$ is a geometric sum with ratio greater than one, proving the second term. Empty classes give no contribution. Later disagreement times are already included in the agreement class through m . $\square$

Proof of Theorem 2.2. The dynamical assertions follow from Lemma 3.1. For a given positive-area compact K , choose $T \subset K$ as in Lemma 5.1. For small δ set

$$m = \min \left\{ n, \left\lfloor \frac{-\log \delta}{\beta(h+\eta)} \right\rfloor \right\}.$$

Then $\delta e^{\beta(h+\eta)m} \leq 1$, and Lemma 5.2 gives

$$\text{area}(T \cap V(n, \delta, u)) \leq 2C_\eta e^{-(h-\eta)m}.$$

Every catalogue for K must cover T , so

$$(5.4) \quad r_G(n, \delta; K) \geq \frac{\text{area}(T)}{2C_\eta} e^{(h-\eta)m}.$$

For $-\log \delta_n/n \rightarrow \gamma$, passage to rates and then $\eta \downarrow 0$ yields the lower bound $\min\{h, \gamma/\beta\}$. Lemma 4.2 and $K \subset Q$ give both upper bounds in (2.7). They coincide with the lower bound when $\gamma \leq \beta h$. The argument includes $\gamma = 0$ and equality at the threshold, and needs no monotonicity of δ_n .

For the high side choose once a compact T_ζ from Lemma 5.1 with area greater than $1 - \zeta$, and set $K_\zeta = T_\zeta \cup \{(0, 0)\}$. For every $\eta > 0$ and all sufficiently large n , all natural prefixes on T_ζ have zero frequency within η of a_0 . The number of these prefixes is at most

$$(5.5) \quad (n+1) \exp \left[n \max_{\substack{0 \leq t \leq 1 \\ |t-a_0| \leq \eta}} H(t) \right].$$

Use each as an exact length- n prefix with any infinite extension, and one extra input for the vertex. This covers K_ζ exactly for all required times, for every $\delta > 0$. Continuity of H gives upper rate h . For every finite $\gamma > \beta h$, (2.7) gives lower rate h on this same set. Its rate therefore exists and equals h simultaneously for all those exponents and all their precision sequences.

It remains to obtain a strictly larger lower rate on Q . Since $a_0 \neq a_1$,

$$\Delta = \chi(1/2) - h = (a_0 - 1/2) \log(a_0/a_1) > 0.$$

For a given $\gamma > \beta h$, let

$$(5.6) \quad \theta_\gamma = \min \left\{ \frac{1}{2}, \frac{\gamma/\beta - h}{2\Delta} \right\}, \quad t_\gamma = (1 - \theta_\gamma)a_0 + \theta_\gamma/2.$$

By affinity of χ and strict concavity of H , $\chi(t_\gamma) < \gamma/\beta$ and $H(t_\gamma) > h$. Choose integers j_n with $j_n/n \rightarrow t_\gamma$. For all sufficiently large n , every word of length n with exactly j_n zeros has

$$-\log \lambda_w = n\chi(j_n/n) < -\log \delta_n/\beta.$$

Every proper prefix has still smaller cost. All these words are therefore terminal leaves of $\mathcal{L}(n, \delta_n^{1/\beta})$. The elementary type bound $\binom{n}{j} \geq e^{nH(j/n)}/(n+1)$ and Lemma 4.2 imply

$$(5.7) \quad r_G(n, \delta_n; Q) \geq \frac{\exp[nH(j_n/n)]}{(n+1)(1+B_0n)}.$$

For completeness, the type bound follows since j is a mode of the binomial law with parameter j/n , so its mass is at least $1/(n+1)$; endpoint types are immediate. Taking lower rates in (5.7) yields $H(t_\gamma) > h$, proving (2.8). The strict gap may tend to zero as $\gamma \downarrow \beta h$. $\square$

Corollary 5.3 (Matched linear systems). *For*

$$A_0(s, z) = (a_0^\beta s, a_0^{\beta-1}(z + a_1 s)), \quad A_1(s, z) = (a_1^\beta s, a_1^{\beta-1}(z - a_0 s)),$$

the conclusions of Theorem 2.2 hold with the same threshold βh .

Proof. Set $\varepsilon = 0$ in the theorem. Then $P_w(s) = \lambda_w$, so the product comparison constant is $D = 1$. $\square$

Remark 5.4. For example, $a_0 = 1/3$, $a_1 = 2/3$, $\beta = 2$ gives $A_0(s, z) = (s/9, z/3 + 2s/9)$ and $A_1(s, z) = (4s/9, 2z/3 - 2s/9)$. Their common-cost threshold is $2[-\frac{1}{3} \log \frac{1}{3} - \frac{2}{3} \log \frac{2}{3}]$. Every admissible positive ε produces a reversible state-dependent system with that same threshold. This is a consequence of the controlled comparison, not merely of the common derivatives at the origin.

6. NO SIMULTANEOUS COORDINATE REDUCTION PRESERVING THE CONSTRAINT

Proposition 6.1. *If $\varepsilon > 0$, there is no ambient homeomorphism J taking Q onto Q and simultaneously conjugating G_0, G_1 to A_0, A_1 , even after permuting the two input labels. In particular, no such C^1 or bi-Lipschitz coordinate change exists. The assertion concerns preservation of Q ; it does not exclude conjugacies that change the constraint set.*

Proof. The states that remain in Q under constant input 0 forever are exactly the lower ray $S_0 = \{(s, -s) : 0 \leq s \leq 1\}$. Indeed, under this input $y + 1$ is multiplied at each step by $p_0(s_k)^{-1} \geq \bar{p}^{-1} > 1$. Every $y > -1$ eventually exits the feasible region. Similarly constant input 1 selects exactly the upper ray $S_1 = \{(s, s) : 0 \leq s \leq 1\}$. The same statements hold for A_i . A conjugacy preserving Q must send these rays to the corresponding target rays, with the possible label permutation, and fix their common vertex.

Let $f_-, f_+ : [0, 1] \rightarrow [0, 1]$ be the radial homeomorphisms on the images of the actual lower and upper rays. Write τ_0, τ_1 for the target radial factors according to the label permutation. Then

$$(6.1) \quad f_- \circ g_0 = \tau_0 f_-, \quad f_+ \circ g_1 = \tau_1 f_+.$$

The dividing state $x_s = (s, s\xi(s))$ admits both controls, and its images are the upper-ray point of radius $g_0(s)$ and the lower-ray point of radius $g_1(s)$. The conjugacy sends the intersection of the two feasible domains to the target dividing ray. If its radius there is $R(s)$, then

$$(6.2) \quad f_+(g_0(s)) = \tau_0 R(s), \quad f_-(g_1(s)) = \tau_1 R(s).$$

These relations hold with either permutation. Set $L_* = g_1 \circ g_0^{-1}$ near zero. Equation (6.2) gives $f_+ = (\tau_0/\tau_1)f_- \circ L_*$ near zero. Substitution into the second relation of (6.1) yields

$$f_- \circ h_1 = \tau_1 f_-, \quad h_1 = L_* \circ g_1 \circ L_*^{-1}.$$

Thus the injective f_- conjugates g_0 and h_1 to two scalar contractions. Those commute, so g_0 and h_1 must commute near zero. This implication uses no differentiability of the proposed conjugacy.

We disprove commutation directly. Write $\rho_i = a_i^\beta$. The explicit maps have expansions

$$g_i(s) = \rho_i s + b_i s^2 + O(s^3), \quad b_0 = \beta \varepsilon a_0^{\beta-1} > 0, \quad b_1 = -\beta \varepsilon a_1^{\beta-1} < 0.$$

Inverting g_0 and composing gives

$$L_*(s) = \frac{\rho_1}{\rho_0} s + \frac{b_1 \rho_0 - \rho_1 b_0}{\rho_0^3} s^2 + O(s^3).$$

For a map $L(s) = As + Bs^2 + O(s^3)$ and $g(s) = \rho s + bs^2 + O(s^3)$, the quadratic coefficient of $L \circ g \circ L^{-1}$ is $b/A + B\rho(\rho - 1)/A^2$. Applying this formula gives

$$h_1(s) = \rho_1 s + \tilde{b}_1 s^2 + O(s^3), \quad \tilde{b}_1 = \frac{b_1(\rho_0 + \rho_1 - 1)}{\rho_1} + \frac{b_0(1 - \rho_1)}{\rho_0}.$$

Consequently

$$(g_0 \circ h_1 - h_1 \circ g_0)(s) = Cs^2 + O(s^3),$$

$$C = (\rho_0 + \rho_1 - 1) \left[\frac{\rho_0(1 - \rho_0)b_1}{\rho_1} - b_0(1 - \rho_1) \right] > 0.$$

Here $\rho_0 + \rho_1 < 1$ because $\beta > 1$, and the bracket is strictly negative. This contradiction proves the proposition. The boundary argument also applies to a locally defined ambient conjugacy whenever it and its inverse are defined on all required one-step images of Q , so feasibility in both directions is preserved. $\square$

7. LOCAL MATCHING DOES NOT SUFFICE

The following example keeps the common attracting origin and even agrees with the linear maps on a neighbourhood of it. It violates global injectivity by collapsing positive-area regions onto an invariant ray.

Proposition 7.1. *For every choice of a_0, a_1, β in (2.2) and every $0 < \sigma < 1/2$, there are C^∞ maps $S_0, S_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with these properties:*

- (1) *They equal A_0, A_1 on $\{s \leq \sigma\}$, so all derivatives at the origin agree.*
- (2) *All inputs drive bounded sets uniformly exponentially to the origin, and Q is controlled invariant.*
- (3) *The compact set $K_\sigma = Q \cap \{s \geq 2\sigma\}$ has area $1 - 4\sigma^2$ and $r_S(n, \delta; K_\sigma) = 1$ for every n and every $\delta > 0$.*
- (4) *For every finite $\gamma \geq 0$ and every sequence $\delta_n \rightarrow 0$ with $-\log \delta_n/n \rightarrow \gamma$, the full-set cost satisfies*

$$\min\{h, \gamma/\beta\} \leq \liminf \frac{\log r_S(n, \delta_n; Q)}{n} \leq \limsup \frac{\log r_S(n, \delta_n; Q)}{n} \leq \min\{\log 2, \gamma/\beta\}.$$

In particular, for $0 < \gamma \leq \beta h$ the full-set rate is γ/β , but the positive-area set K_σ has rate zero.

Proof. Let $e(t) = 0$ for $t \leq 0$ and $e(t) = \exp(-1/t)$ for $t > 0$, and define

$$\psi(s) = \frac{e(s - \sigma)}{e(s - \sigma) + e(2\sigma - s)}.$$

The denominator is positive everywhere. Thus ψ is smooth, zero for $s \leq \sigma$, and one for $s \geq 2\sigma$. Put $\rho_i = a_i^\beta$ and define

$$(7.1) \quad S_0(s, z) = (\rho_0 s, (1 - \psi(s))a_0^{\beta-1}(z + a_1 s) - \psi(s)\rho_0 s),$$

$$(7.2) \quad S_1(s, z) = (\rho_1 s, (1 - \psi(s))a_1^{\beta-1}(z - a_0 s) + \psi(s)\rho_1 s).$$

They have the claimed local identity. Each $S_i(s, z)$ is a convex combination of $A_i(s, z)$ and $R_i(s, z)$, where $R_0 = (\rho_0 s, -\rho_0 s)$ and $R_1 = (\rho_1 s, \rho_1 s)$. All four linear maps contract toward zero in one adapted norm $\max\{M|s|, |z|\}$ for sufficiently large M : the second-coordinate factor for A_i is at most $a_i^{\beta-1} + a_i^{\beta-1}a_{1-i}/M < 1$, and R_i has factor at most ρ_i . Convexity of the norm gives the same contraction bound for S_i on all of $\mathbb{R}^2$.

For controlled feasibility let $t = \psi(s)$, $y = z/s$, and $\xi = a_0 - a_1$. If $t < 1$ the full feasible intervals are

$$J_0(s) = \left[-1, \min \left\{ 1, \xi + \frac{2a_0 t}{1-t} \right\} \right],$$

$$J_1(s) = \left[\max \left\{ -1, \xi - \frac{2a_1 t}{1-t} \right\}, 1 \right].$$

They follow by imposing $-1 \leq z'/s' \leq 1$ in (7.1)–(7.2), and their union is $[-1, 1]$. For $t = 1$ both branches are feasible on the whole fibre. The radius always decreases, so Q is controlled invariant.

On K_σ , one application of S_0 sends every state to the lower ray. At every radius $S_0(s, -s) = (\rho_0 s, -\rho_0 s)$, so input 0^∞ keeps the whole set exactly in Q . Its area is $\int_{2\sigma}^1 2s ds = 1 - 4\sigma^2$.

To obtain the full-set lower bound, use $Q_c = Q \cap \{s \leq c\}$ with $c = \sigma/2$. For every input, the radius of a state in Q_c stays at most c . Thus $\psi = 0$ forever along that trajectory, even when its angle leaves $[-1, 1]$, and its entire trajectory is exactly that of A_i . Consequently $r_S(n, \delta; Q) \geq r_A(n, \delta; Q_c)$. Corollary 5.3 and (2.7) give the lower bound.

For the upper bound choose an integer t_0 such that $(\max_i \rho_i)^{t_0} \leq \sigma$. Every initial state admits an exact feasible prefix of length t_0 , ending in the region $s \leq \sigma$. All subsequent trajectories there coincide with the linear system, for all inputs, since the radius continues to decrease independently of the angle. Concatenate all 2^{t_0} possible prefixes with a catalogue for the linear full set over the remaining horizon. For $n \geq t_0$ this gives

$$r_S(n, \delta; Q) \leq 2^{t_0} r_A(n - t_0, \delta; Q) \leq 2^{t_0} C_U \delta^{-1/\beta}.$$

The 2^n exact-prefix bound also holds. These yield the upper rates for every stated precision sequence. Finally $\det DS_i = (1 - \psi(s))a_i^{2\beta-1}$ vanishes on the saturated region, identifying the noninjective mechanism. $\square$

For fixed σ this proposition gives one large zero-cost initial set in one fixed system. Making its area arbitrarily close to one changes σ and the system. This differs from Theorem 2.2, where all K_ζ are chosen inside a single fixed system. The collapse example is a counterexample to sufficiency of local matching and uniform convergence; it is not a counterexample to the reversible theorem.

8. SCOPE AND CONSEQUENCES

The low-precision law uses positive area without requiring interior. Its proof remains valid near the vertex because the singular factor $1/s$ in the first-deviation band is cancelled by the physical area element. Above the threshold, typical compact sets have rate h , while atypical terminal words force a larger full-set lower rate. The threshold therefore concerns the quantifier over *all* positive-area compact initial sets, rather than uniform convergence of the controlled states.

The state-dependent result preserves the exact relation between angular branch length and radial contraction, but removes the requirement that branches be fixed linear intervals. Uniform distortion alone is not used as a substitute for actual control feasibility: the comparison of arbitrary inputs with stopped physical witnesses is what permits counting estimates to govern the true catalogue. Proposition 6.1 distinguishes this extension from a change of coordinates preserving the original constraint, and Proposition 7.1 shows a concrete obstruction outside the invertible class.

Several limitations remain. The systems are triangular, the instantaneous matching is exact, and the perturbation has a specific summable state dependence. We prove no general nonlinear perturbation theorem based solely on invertibility or the derivative at the origin. For the full set above the threshold, the strict lower bound (5.7) need not match the upper bounds in (2.7); a general exact spectrum is not claimed. The main theorem

instead completely resolves the common-cost boundary and its simultaneous initial-set quantifiers in the stated family.

Generative AI disclosure. Generative AI was used extensively in developing mathematical arguments, checking proof chains and related literature, drafting and restructuring the English text, and producing the LaTeX source. These activities do not constitute verification by a human author. Author identity, affiliations, authorship approval and the scope of human mathematical review remain to be supplied and confirmed before any submission.

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